

PROJECT REPORT

WhatsApp AI Agent

Smart Business Assistant using n8n + Groq AI + Google Sheets

1. Introduction

This report documents the successful design, development, and testing of the WhatsApp AI Agent proposed earlier in the project proposal. The agent is a fully automated business assistant built on the n8n workflow automation platform, integrating Groq's LLaMA AI model, WhatsApp Business Cloud API, and Google Sheets. It allows customers to browse inventory, ask FAQs, and place orders directly through WhatsApp, with every conversation and order handled automatically without human involvement.

The following sections describe how the workflow was implemented, the real challenges encountered during development, the solutions applied to resolve them, and the final testing results confirming that the agent functions correctly end-to-end.

2. Implementation Summary

The workflow was implemented in n8n Cloud following the steps outlined in the proposal:

- Set up an n8n Cloud account and created a new workflow named 'My workflow'.
- Created three Google Sheets — Inventory, FAQ, and Orders — and connected them to n8n using Google Sheets OAuth2 credentials.
- Obtained a Groq API key from console.groq.com and configured the Groq Chat Model node using the LLaMA 3 model.
- Built the AI Agent node and connected it to four tools: Simple Memory (conversation history), Get Inventory (reads Inventory sheet), Get FAQ (reads FAQ sheet), and Post Orders (writes to Orders sheet).
- Configured a WhatsApp Trigger node to receive incoming customer messages and a Send Message node to reply back on WhatsApp.
- Wrote a system prompt defining the agent's role, tone, and rules for handling menu queries, FAQs, and order placement.

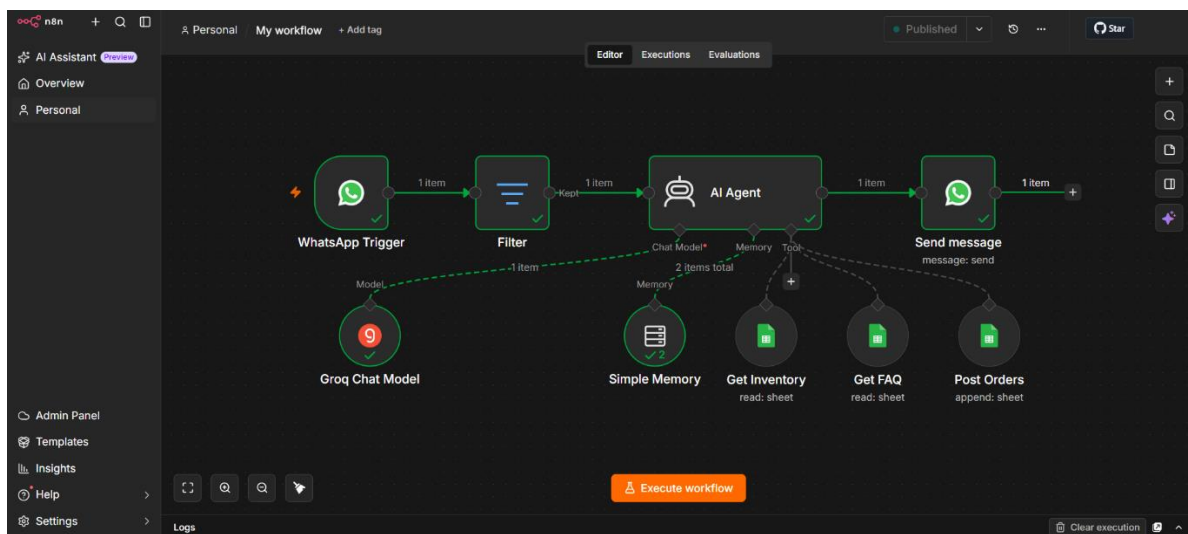


Figure 1: Complete n8n workflow architecture — WhatsApp Trigger, Filter, AI Agent (with Groq model, Memory, and Sheet tools), and Send Message.

3. Tools & Technologies Used

Tool / Technology	Role in Project	Node in n8n
n8n Cloud	Core workflow builder and automation engine	All nodes
Groq AI (LLaMA 3)	AI language model to understand and respond to messages	Groq Chat Model node
WhatsApp Business Cloud API	Channel for receiving and sending customer messages	WhatsApp Trigger + Send Message
Google Sheets	Database for Inventory, FAQ, and Order records	Get Inventory, Get FAQ, Post Orders
Simple Memory	Stores conversation history for multi-turn chat context	Simple Memory node
AI Agent (n8n)	Orchestrates all tools and decides which action to take	AI Agent node

4. Challenges Faced & Solutions

During development and testing, several real-world issues were encountered and resolved. Documenting these was an important part of understanding how production AI workflows behave outside of a controlled tutorial environment.

Challenge 1: Groq API Rate Limiting

While testing the AI Agent repeatedly, the Groq Chat Model node returned the error "The service is receiving too many requests from you". This occurred because the free tier of the Groq API has a limited number of requests allowed per minute.

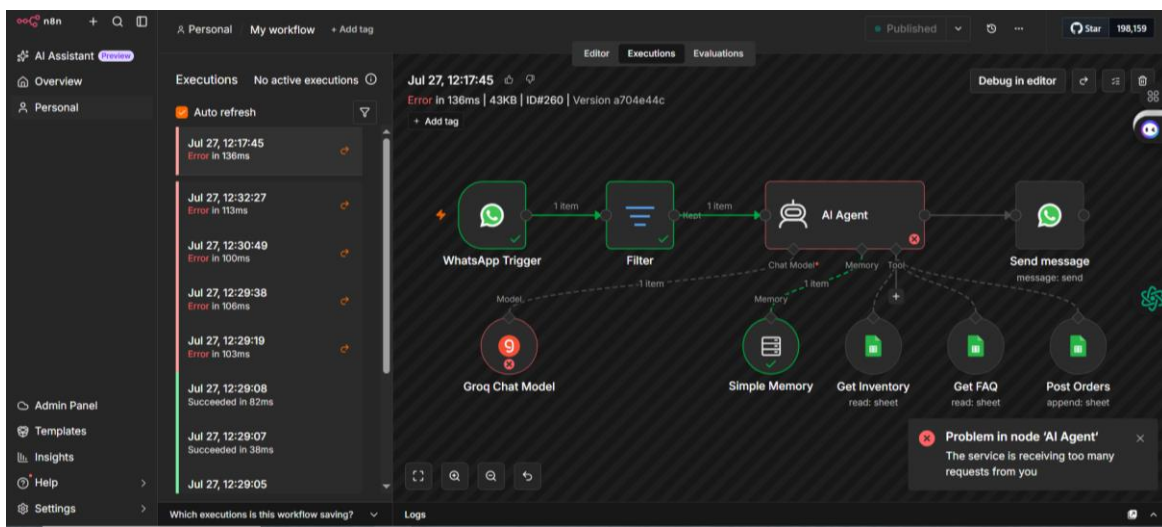


Figure 2: Rate limit error returned by the Groq Chat Model node during repeated testing.

Solution: Testing frequency was reduced, and a short delay was introduced between consecutive executions so that requests stayed within the free-tier limit.

Challenge 2: WhatsApp Access Token Authorization Failure

The Send Message node began failing with "Authorization failed — please check your credentials". This happened because the temporary WhatsApp access token issued by Meta expires automatically after 24 hours.

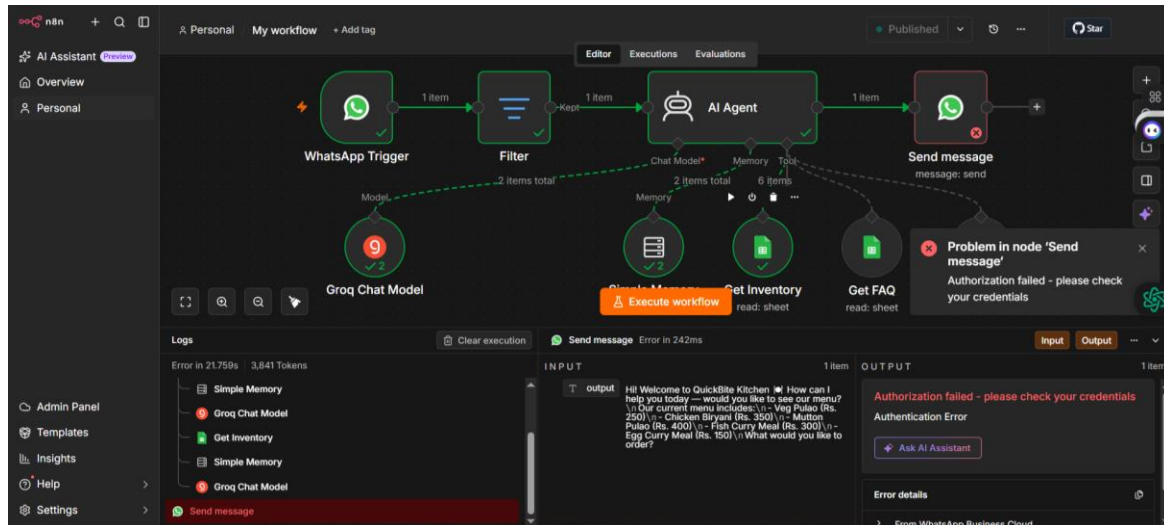


Figure 3: Authorization error caused by an expired WhatsApp access token.

Solution: A new access token was generated using Meta's Graph API Explorer with the `whatsapp_business_messaging` and `whatsapp_business_management` permissions enabled, and the credential in n8n was updated accordingly.

5. Testing & Results

After resolving the above issues, the complete workflow was tested by sending real messages through WhatsApp, including menu queries and order placement. The workflow executed successfully end-to-end multiple times, confirming stable and repeatable behaviour.

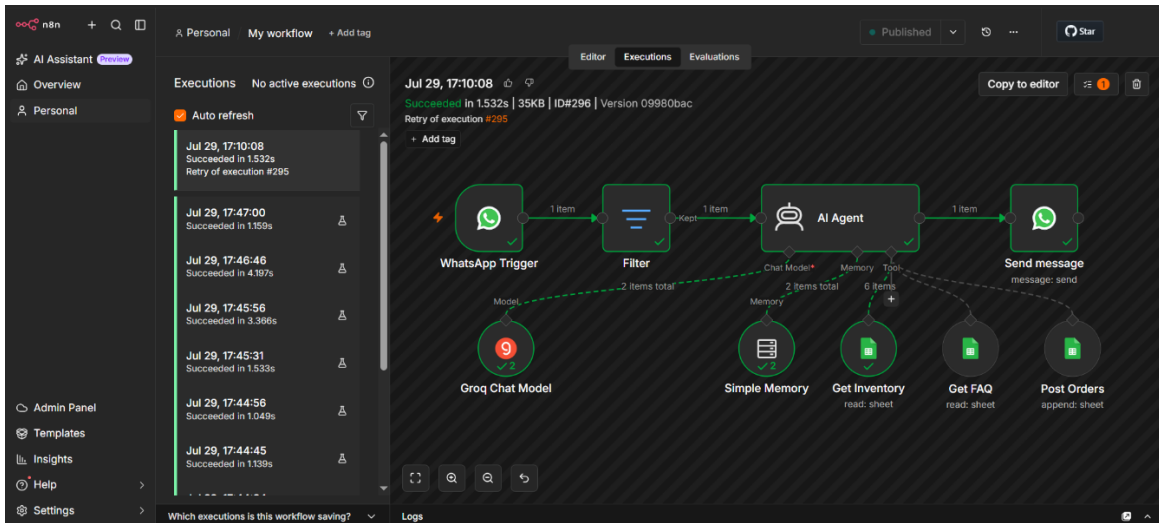
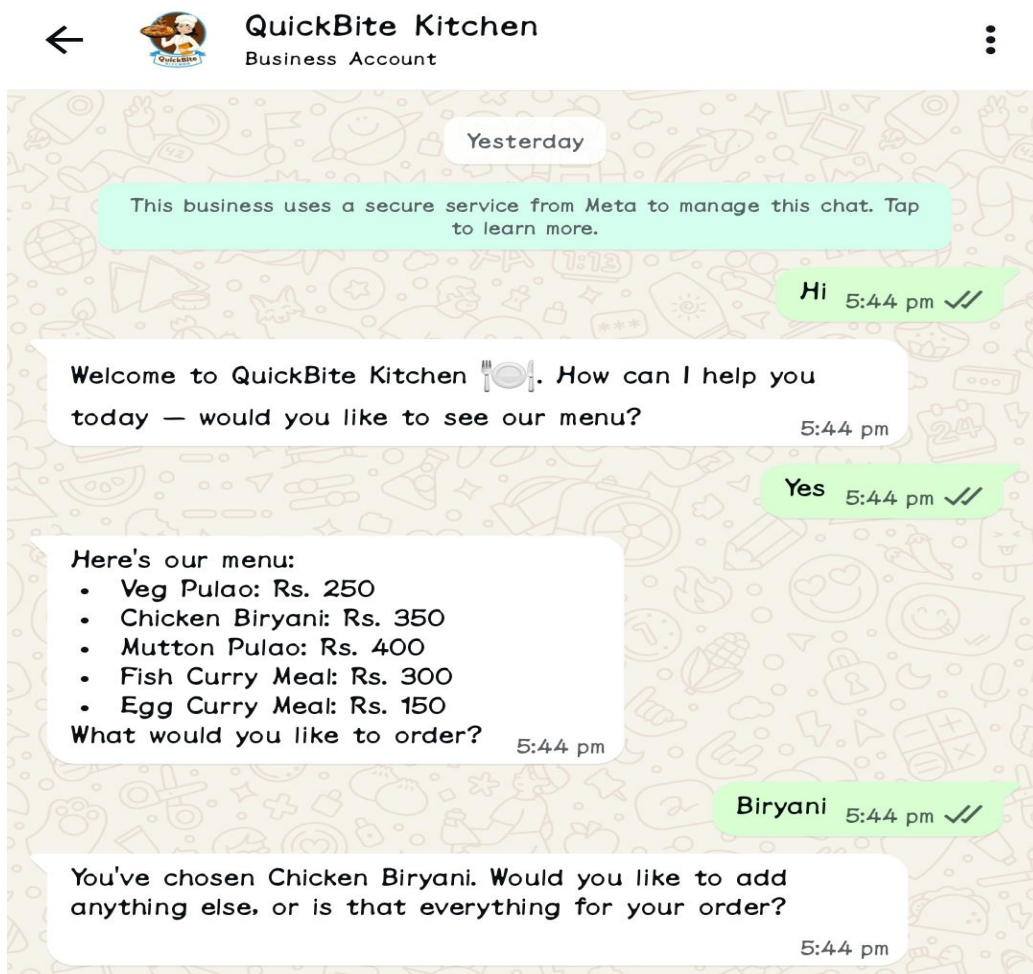
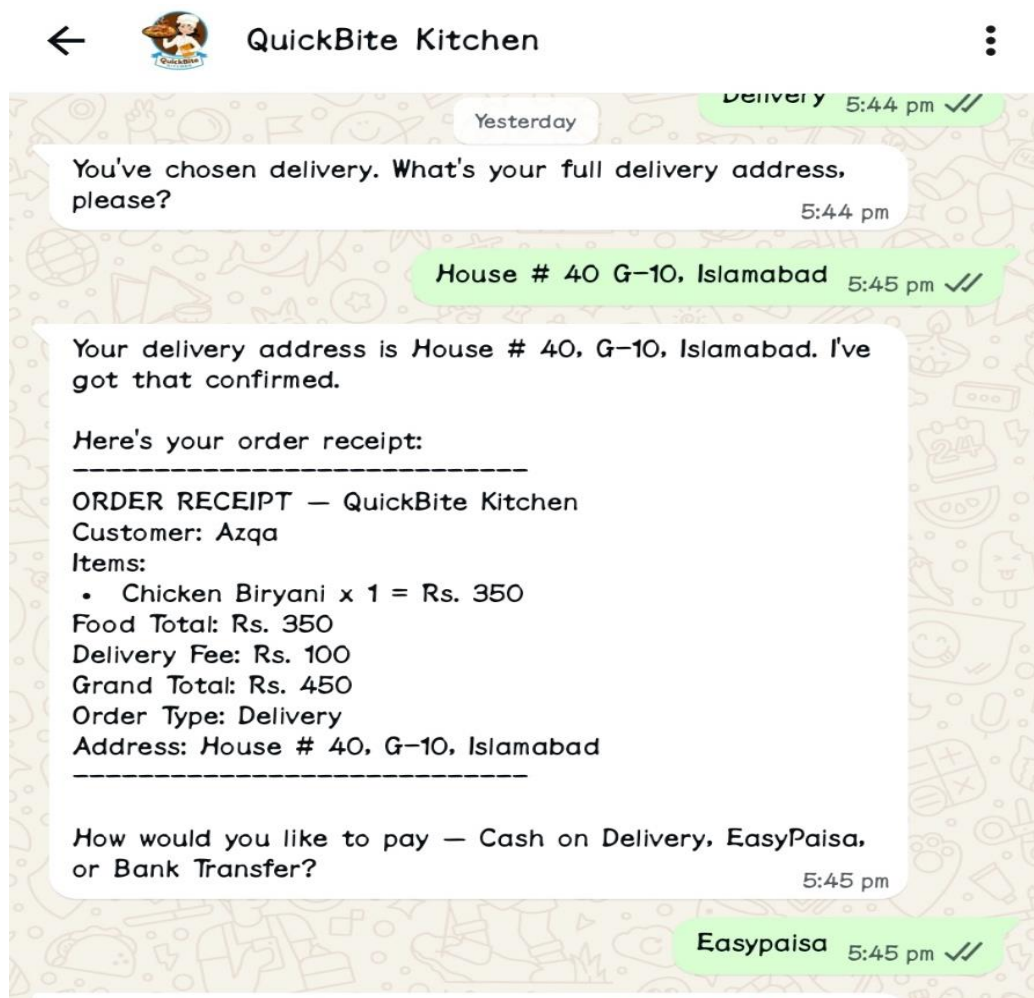
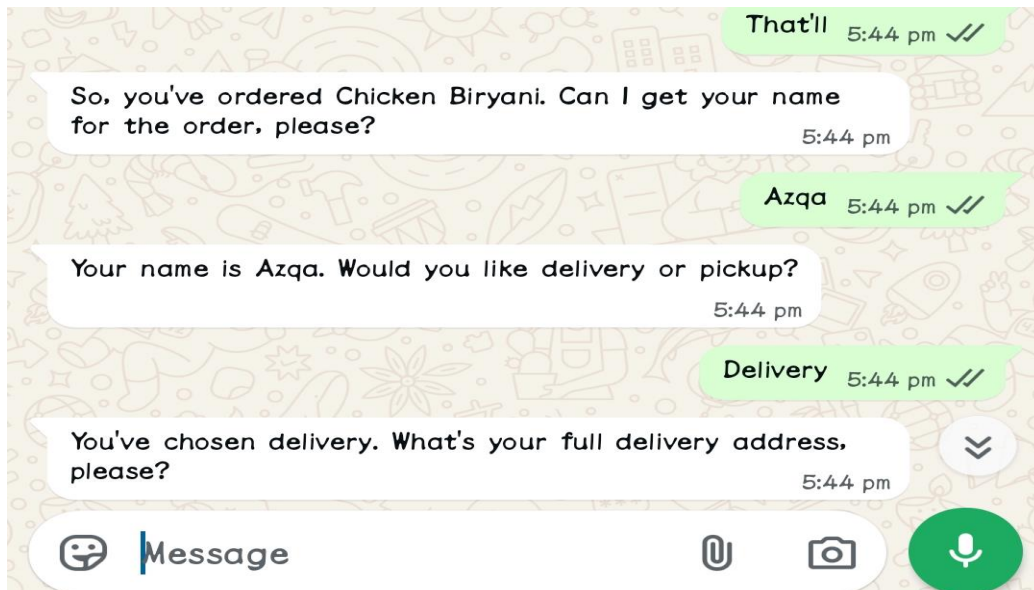


Figure 4: Execution history showing multiple successful workflow runs.

To confirm the agent behaves correctly from the customer's side, a real conversation was carried out on WhatsApp — asking about the menu and placing an order — and the AI Agent's replies were verified in real time.





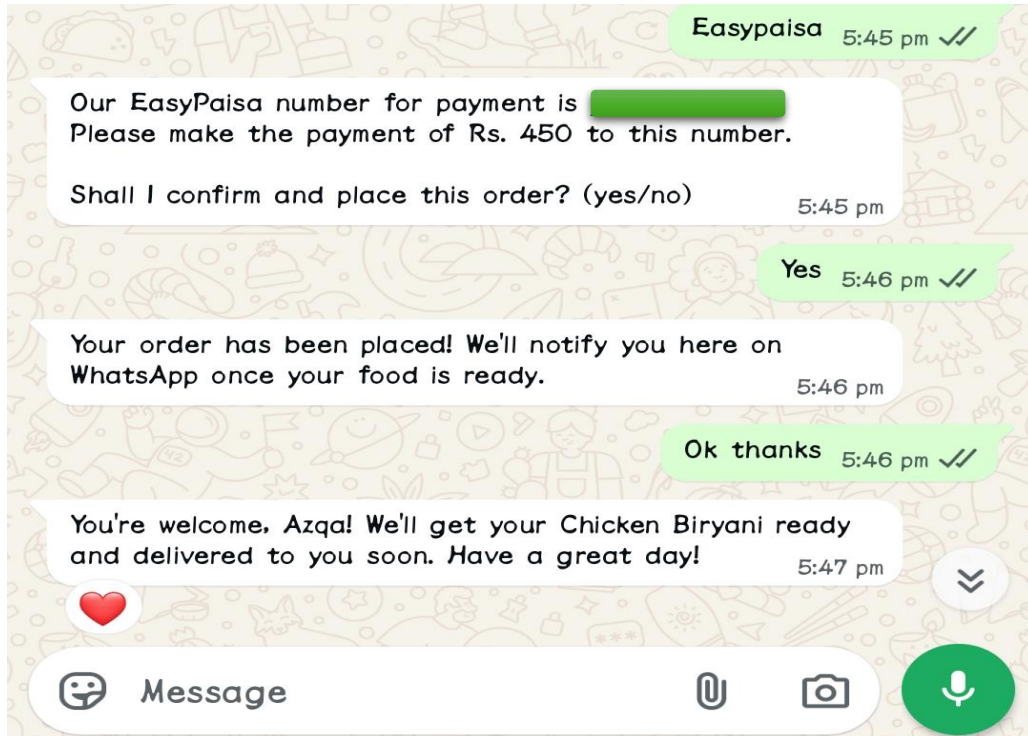


Figure 5: WhatsApp conversation showing a real customer order being placed and confirmed by the AI Agent.

Orders placed through WhatsApp were verified to be logged automatically into the Google Sheets Orders tab, without any manual data entry, confirming that the Post Orders tool functioned correctly.

The image shows a Google Sheet titled 'Food Delivery System'. The 'Orders' tab is selected. The sheet contains the following data:

Customer Name	Food Items	Quantity Ordered	Order date	Status	Order Type	Delivery Address	Payment Method	Total Amount
Azqa	Chicken Biryani	1	2026-07-29 05:46:48 PM	Ordered	Delivery	House # 40, G-10, EasyPaixa		450

Figure 6: Order automatically recorded in the Google Sheets Orders tab after a customer placed an order via WhatsApp.

6. Conclusion & Future Scope

The WhatsApp AI Agent was successfully built, tested, and verified to work as intended. It automatically answers menu and FAQ questions using live Google Sheets data, takes customer orders conversationally, maintains context across multiple messages using Simple Memory, and

logs every confirmed order without manual intervention. Real issues such as API rate limiting and access token expiry were encountered and resolved, giving practical experience in maintaining an AI-driven automation in a production-like setting.

Future improvements could include:

- Replacing the temporary WhatsApp access token with a permanent System User token to avoid periodic re-authentication.
- Migrating Inventory, FAQ, and Orders data from Google Sheets to a proper database for better scalability.
- Adding payment integration for a fully self-service ordering experience.
- Extending the agent to handle order status updates and delivery tracking.